

Deployment and support of complex integrated solutions

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BT needs to bring new products and services faster into the market-place. Consequently, network and system components must be deployed more rapidly than ever. Successful integration and launch processes demand efficient, effective delivery and commissioning of new components followed by equally efficient day-to-day operational support. This paper discusses the support framework needed to satisfy these demands.

1. Introduction

Very few companies integrate new information systems into a complex telecommunications network as large and complex as BT's. The scale of the task means that no single group of people can undertake all aspects of the work involved in the delivery and deployment process. New products and services are often introduced as part of a programme of work. The programme owner has the responsibility for creating the processes needed to deliver business solutions while ensuring that an appropriate support infrastructure is in place.

A key requirement for success of the integration-led approach is the harnessing of complementary skills across many organisational boundaries. This may require the rapid acquisition of new skills where they do not currently exist. The ultimate aim is to deliver new products or services into 'business-as-usual' processes as quickly as possible. BT already has a well-established planning, deployment and support infrastructure that deals with the day-to-day operational aspects of service delivery, performance and capacity planning.

The challenge facing those responsible for the deployment and support of new products and services is to merge new functionality with existing functionality. The aim is to gain maximum business benefit from rapid service introduction. Similarly there is a requirement to ensure that an appropriate infrastructure is in place to support the changes in ongoing operation.

The validation techniques employed to ensure that the new functionality works, and that it does not adversely impact on existing services are described elsewhere [1]. This paper addresses the issues surrounding the deployment and technical support of the computing platforms and systems that constitute these complex integrated solutions.

2. Deployment and support in the product life cycle

The product life cycle is shown in Fig 1. This paper addresses the delivery, installation and customer support stages.

Customer acceptance and sign-off are achieved at the point of hand-over into operations and hence ongoing support. This may be a separate activity or combined with a pilot trial. Pilot trials offer the opportunity to test the operational effectiveness of the system and the service surround prior to a wider roll-out.

The objective of the delivery and installation phase is to provide the customer with a working product within the agreed time-scales. The user also requires the product to be fit for purpose. The main activities of delivery and installation occur towards the end of the life cycle, but the planning and preparation usually start much earlier in order to meet procurement lead times.

The key stages in this part of the life cycle are discussed in section 3 below.

The support activity, sometimes called operations and maintenance, is the phase that deals with problems in operational service. It is important at this stage to understand the difference between support for the system and maintenance. A simple analogy is to compare support and maintenance with the services offered to motorists by a motoring organisation, such as the AA, and garages. When a car breaks down in use, the customer makes a call to the AA. An AA patrolman answers the call and takes the necessary action to enable the motorist to complete their journey. This action may be a temporary repair or the fitting of a replacement part. If a more serious problem is found, or the problem re-occurs, the car is taken to the garage for in-depth analysis and repair. In some cases the garage will need to refer the problem back to the original manufacturer.

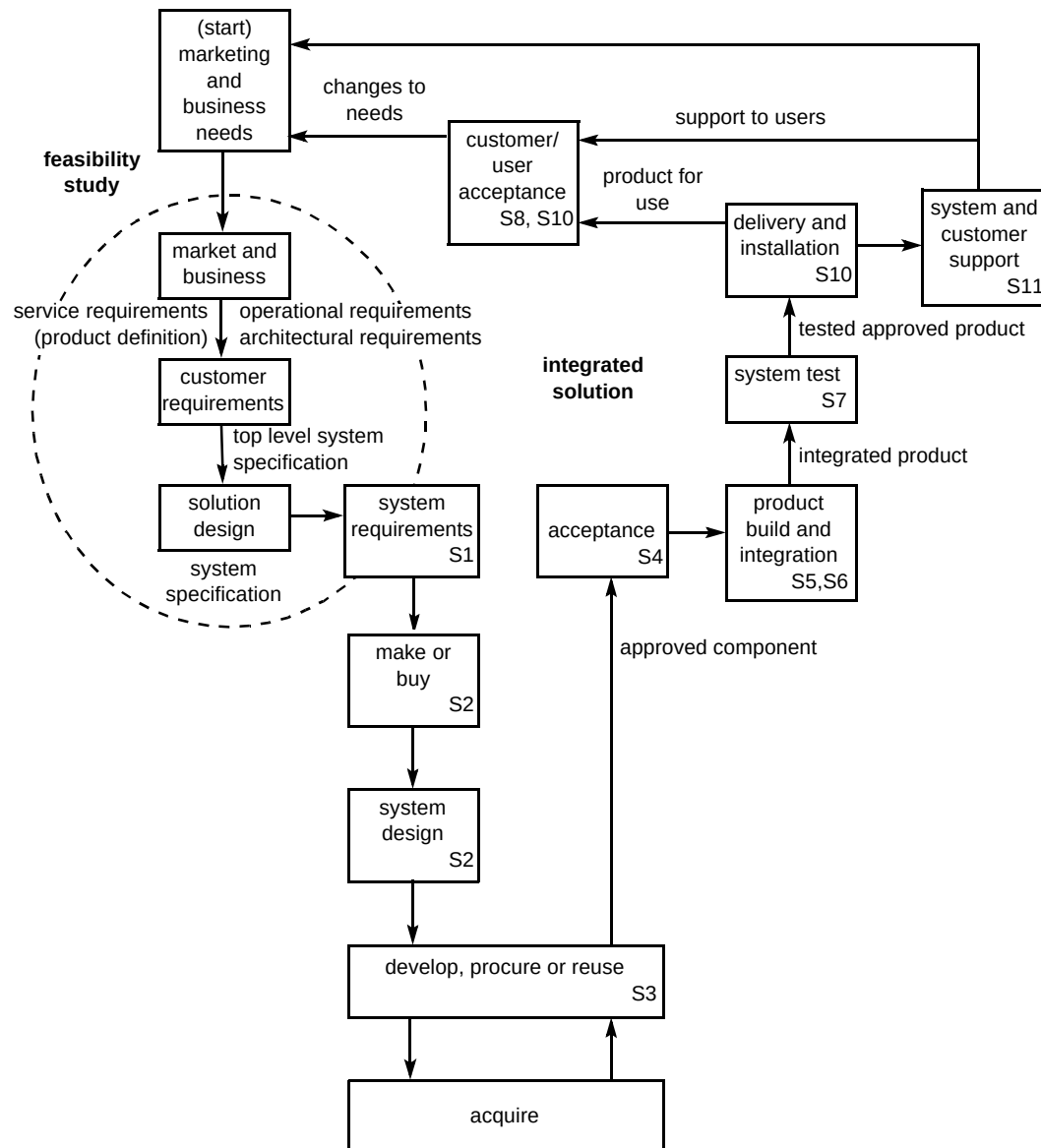


Fig 1 The product life cycle.

Operational support for a complex system is similar to the service offered by the AA. When the system fails to perform as expected a call is made to the support organisation. Following investigation by a support engineer a temporary fix may be implemented by changes to data or configuration. If a more serious problem is found, such as a bug in the application code, then the problem is referred back to the supplier to provide a fix. This provides the equivalent service of the garage or the manufacturer. Although the analogy is useful it only goes part of the way in describing the role of in-service support. Section 4 describes the support process in more detail.

3. The delivery and installation process

This process co-ordinates the delivery, installation and commissioning of formally released components

(hardware, software, etc) into target environments. Successful completion of pre-agreed tests are also demonstrated to the customer at this point.

The principle of 'first verification office' (FVO) is normally applied. Here, the delivery/installation team limits their involvement to the early deployments and focuses their expertise to overcome any initial difficulties and problems.

FVO includes all the planning, preparation and documentation activities required to deploy the product within the given time-scales. Entry and exit from this process are controlled by quality gates. Figure 2 shows the stages involved in this process.

Quality gates provide a structured hand-over between the various parts of the business. They also define internal

milestones for the detailed planning work that helps to provide management control.

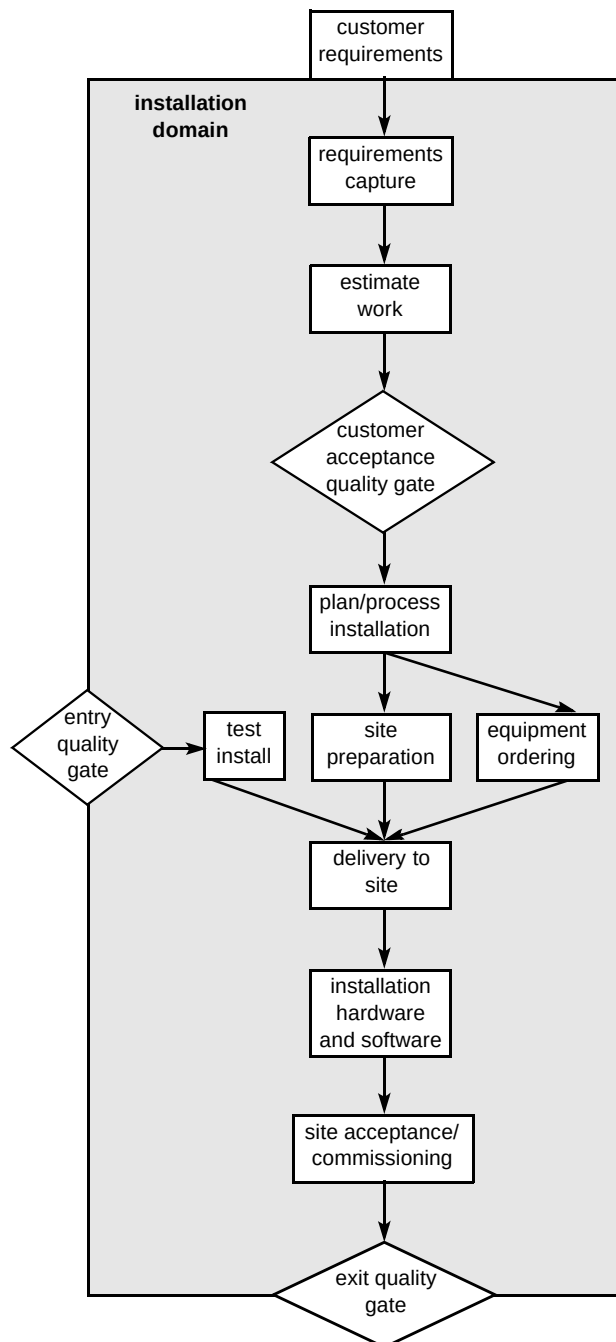


Fig 2 Outline installation process.

A 'customer acceptance' quality gate defines the delivery agreement. The detail of the quality gate is determined in discussion with the customer prior to hand-over. The agreement would typically cover what is to be done, by whom and in what time-scales.

The quality of discussion at this point is vitally important as it helps to manage expectation and ensure that all aspects of the delivery are taken into account. Details

such as the scope of acceptance tests, responsibilities for equipment provision, and site documentation are usually covered.

With complex deliveries involving many external and internal suppliers, it is important that activities with long lead times such as equipment ordering are addressed early. Building long-term relationships with suppliers can assist this process and shorten lead times. Often equipment ordering takes place in parallel with site preparation.

A local site preparation meeting will further define the responsibilities of the customer, equipment suppliers and deployment teams. Accommodation issues such as power, ventilation, alarms, and furniture would normally be covered also, ensuring that any potential access or security difficulties are eliminated.

The 'entry gate' authorises the release of the product to be installed. 'Hot staging' before delivery to site is used to validate the installation guide, delivered media and target hardware and software.

At the time of delivery items are checked against the customer requirement's delivery schedule. Site details are also checked including hardware layout/configuration, power, communications, licences, and documentation.

During installation any faults or difficulties that arise are reported to the appropriate owners and managed through by the deployment team. Depending upon the type of delivery a series of pre-agreed site acceptance or commissioning tests may be run and witnessed by the customer.

To achieve sign-off a delivery will require a minimal set of tests to demonstrate major functionality. Any necessary corrective actions are agreed with the customer or handed to the support agency responsible for resolving them.

The 'exit gate' signifies the end of the installation phase and formalises the hand-over to the next stage owner, usually the customer support organisation. At this stage the formal customer acceptance certificate is signed.

4. Delivering support

4.1 Support consultancy

New products and services are designed to meet customer's expectations regarding availability, reliability and mean time between failure criteria.

Many system designs, however, underestimate the complexity of deploying large new solutions into an existing network. Similarly they make insufficient allowance for cost-effective operational maintenance.

Operational considerations typically may be the time taken for system restoration following major component failure, or the time it takes to restore and align a large database.

The reliability of complex operational systems is generally less than that predicted at the design stage. A well-designed and implemented maintenance and support framework can make a significant contribution to cost effectiveness and the overall system availability.

Networks and their management systems are becoming more reliant upon software; hence minor errors within code may pose considerable risk to the operation of the systems involved. The modified Richter scale in Fig 3 [2] seeks to indicate the impact operational failures may have on an individual, company, even the nation.

failure type	Richter scale	range of effects
local network failure switch node failure national network control international network control	10	major economic disruption
	9	
	8	government concern and action
	7	
	6	national press reports
	5	
	4	local press reports
	3	
	2	individual complaints
	1	within normal conduct

Fig 3 'Richter' outage disaster scale for networks [2].

Support organisations are constantly faced with resolving issues and problems that occur when systems or components do not work as predicted.

Consequently designers and product owners often look to support for guidance with regard to achieving maximum availability of systems and how to minimise the impact of failure. Support consultancy can also assist with defining service restoration under different failure scenarios.

Consultancy creates an opportunity early in the life cycle for experienced support people to join project teams working to achieve these aims. Similarly the programme may appoint a sole owner for the delivery of in-service operational requirements and support infrastructure. Issues considered during this initial consultancy include:

- definition and agreement of support process,
- definition of support procedures and responsibilities,
- reuse of agreed operability standards,
- tools and facilities required,
- negotiation of service level, and technical service, agreements with customers and suppliers,
- the cost of support.

Support consultants can assist in aligning operational requirements with design specifications via their ability to understand the language used in the operations community and that of design and development.

A further benefit of using experienced support people at this stage is to achieve economies of scale, short evolution times and robust support processes. Integrated support strategies that employ best-practice can be defined rather than each new programme having to design yet another bespoke support organisation.

Using support consultants early in the project life cycle removes many obstacles to achieving fully operational systems at the earliest possible opportunity. Unplanned or unclear support strategies have a consequential risk of greatly extended repair times. Given the impact of system or service failure, removing or minimising this risk will bring benefit to the business.

4.2 Technical support framework

As the number of network and support systems within BT has grown, the demand for technical support with its attendant processes has grown proportionately.

Business relationships can be impeded by differing interpretations of the definition of 'support'. Communication within the company and that with outside suppliers has also exposed differing interpretation of commonly used words and terms, such as expert, first line support, etc.

Due to the reoccurrence of common themes, and despite the variety of terms used, it is evident that the support process can be described by function and activity, rather than using colloquial terminology such as 'second line' and 'ownership'. Most support frameworks align with that shown in Fig 4.

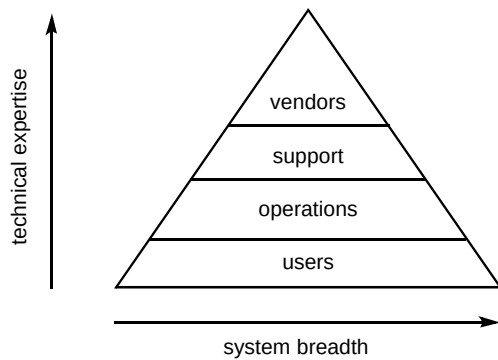


Fig 4 The support framework.

‘Users’ generally have the greatest breadth of knowledge of the wider business process. ‘Operations’ are responsible for delivering the service against defined service targets.

If a problem cannot be resolved locally then operations seek assistance from the support organisation. The support organisation will be expert in one or more of the component parts of the system. Depending on the scale and the system, the support organisation could be multi-layered.

Hardware and software manufacturers are referred to as vendors. Vendors may supply complete systems or components that contribute to the overall product or service. Major vendors include DEC, SEQUENT, HP, SUN, Compaq, CISCO, etc, as well as software-only manufacturers such as BT, Microsoft, and Oracle.

The designer of an integrated solution using off-the-shelf components can also be considered as a vendor. In some cases, vendors might also be contracted to provide hardware repair services. In almost every case the vendor will be characterised by an in-depth knowledge of the component, having been heavily involved in its original production.

4.3 Responsibility domains

In any organisation, particularly one as large as BT, the logical model of Fig 4 has to be translated into a practical implementation that is generally bound by organisational constraints. This requires responsibilities to be assigned to the most appropriate place. Figure 5 shows the total support service split into four domains, each of which is characterised by a different focus and skill set.

- Customer centric — the customer centric domain embraces all aspects of the service as seen by the customer or end user. This domain defines the service standards and may also be the primary point of contact for customers.

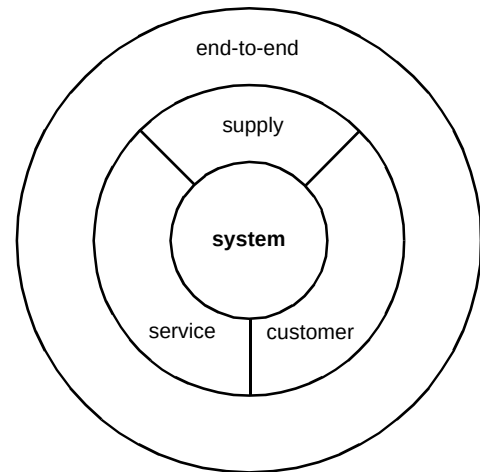


Fig 5 Support service domains.

- Service centric — the service domain is responsible to the customer domain for the delivery of the technological service required, i.e. computing and communications. The focus is on the service standards required of the technology components and advising on best practice with regard to optimising product performance.
- Supply centric — the supply domain delivers components, designs and component repairs into the service domain. It has responsibility for meeting agreed standards for the delivery, replacement, or repair of installed components. Reactive focus is required on any requests for funded redesign and bug-fixing activities. Known problems with the delivered components are advised to the service domain. Build control is authorised and baseline configurations are maintained.
- End-to-end service ownership — this responsibility refers to the service as seen by the end user or customer. As the service might be provided via a range of different products there could be multiple operations and support organisations contributing to the overall service. All service and vendor centric organisations need to ensure that they recognise the focus, responsibility and authority of the end-to-end service owner. Clear ownership of this role by a suitably experienced and informed person or persons is essential if the service as seen by the end user or customer is to be optimised.

4.4 A practical support model

An example support model is shown in Fig 6. The roles are described by what they do rather than any particular name or term. Multiple roles may be performed by a single team.

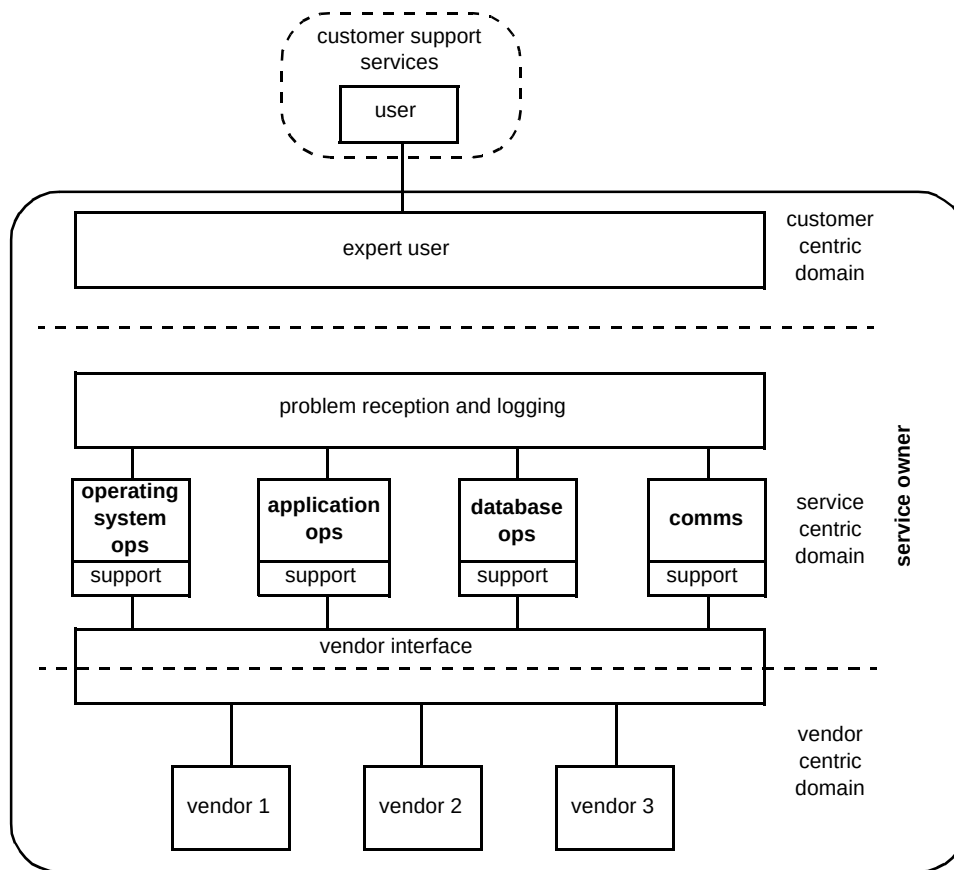


Fig 6 A practical support model.

- Expert user or user helpdesk (first point of contact by users) — this role will be filled by an individual who has an above average knowledge of the product or by a team who are dedicated to dealing with customers' queries on a full-time basis. The key activities of this role are typically to resolve queries related to the product or service usage, maintain ownership of user queries until completed, maintain a log of calls dealt with (for tracking and metrics purposes). They will be able to categorise calls into those that require technical support from those related to business processes such as billing or ordering.
- Problem reception and logging — a problem logging capability is important for an understanding of the service provided across numerous interconnected products. This role may also provide initial diagnostics, and an awareness of service downtime, and ensures consistent routing, logging and tracking of the problem and service-related reports. This role will be the central source of service information to both the customer and service provider with problem tracking, metrics against service level agreement, and problem status updates.
- Dealer groups — the number of specialist dealer groups required for a given product or service will vary from project to project dependent on the component architecture. For reasons of practicality and economy,

the role of multiple dealer groups may be taken on by a single multi-skilled team. One dealer group must have expertise in system-wide fault diagnosis and be formally identified as such to the 'Problem reception and logging' team. Where economies of scale exist, specialisms may be split into different levels of expertise, e.g. operations and operations support, or platform operations and platform operations support. Dealer groups must ensure that all problems assigned to them are either cleared or passed to a more appropriate group. 'Problem logging and reception' must be kept advised of progress with regard to assigned faults.

- Vendor interface — this dealer group is particularly important for complex system integration tasks, because it provides the link across all specialisms within the product or service. They will understand the functionality of all the components, their interfaces and service requirements at a sufficient level to be able to diagnose and resolve end-to-end service issues. Release management for patches and new builds will be provided, concentrating on system compatibility issues. They will take the lead in investigating and resolving the complex problems related to end-to-end performance and provide a cross-programme view of problems passing between vendors and operational

support areas. The team will maintain knowledge of all key contacts in programme, customer, vendor and operational support areas relating to the project.

- Vendor — software developers, integrators and suppliers, computer and communications hardware manufacturers and external maintenance organisations can be considered as vendors. They provide expert back-up to operational support, supplying bug fixes to application software, owning application source code and providing configuration management of their components.

4.5 Launch support

During the initial launch of a new product or service it may be appropriate to have a single team that performs all or any subset of the roles described above. The objective of this team is to provide all-encompassing support during the initial launch and operational trials of the system, when the likelihood of problems is greatest. If the trial is successful and long-term support is required, then the launch support team also has responsibility for moving the product into the mainstream support infrastructure. This will involve providing assistance in developing operational procedures, skills transfer and implementation of the support processes.

5. Conclusions

In summary, delivery and support are as important as the initial design and manufacture for the successful implementation of BT's key operational systems.

As BT increases its focus on customer services, the requirement will be to deploy new products and services more rapidly. Globalisation, mobility, new network technologies, entertainment and information services will drive the demand for systems to be integrated into the existing network and support structures. Technology advances will enable greater flexibility, reliability and rapid introduction of new functionality.

All of these changes will require a nimble and forward-looking deployment and support organisation that can meet these challenges while continuing to provide cost-effective and professional support to the existing systems.

Acknowledgements

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Ian Buxton joined BT in 1972 as an apprentice. He graduated in 1977 with a degree in electrical engineering from Aston University. After a number of years lecturing in an engineering training centre he moved to BT Laboratories to work in software development, implementing interactive services on the Westminster Switched Star Cable TV network. He then headed a number of teams providing technical support for some of BT's key operational support systems including the operations and maintenance centre (OMC2), Work Manager and Serviceview manager.

He currently manages a unit in Systems Integration department providing operational and deployment support. He is an associate member of the IEE.



Steve Hodgkiss joined BT in 1972 as an apprentice in the East Midlands. After a period of involvement in electromechanical exchange maintenance, he moved into commissioning and maintenance of digital exchanges. He has a BA from the Open University and a Diploma in management studies from Nottingham Trent. In 1986 he joined the operations and maintenance centre product support team at BT Laboratories working on the basic OMC and subsequently OMC 2 projects.

He currently leads a group whose activities include software release management and operational support for the SDH and DESS operational support systems.



Geoff King joined BT as a trainee technician in 1968. Since then he has enjoyed a varied career, mainly based in London, which included exchange construction, transmission planning, lecturing at the engineering training centre and technical support for the operation and maintenance of digital exchanges.

In 1987 Geoff moved jobs to Ipswich, taking on responsibility for rationalisation of computer centres to form the network administration computer centres (NACCs). Installing systems like OMC2 (operations and maintenance centre) formed the backbone of the ongoing work, but this has extended over the years to cover a multitude of different systems and platforms. He is currently responsible for a team of people, collectively known as Systems Deployment.